

Base R Graphics — devices & par() cheatsheet

Data Visualization with R · viz-base.rsquaredacademy.com · CC BY-NC-SA 4.0

DEVICES

Devices are a stack. Opening one makes it active; closing it returns to the one underneath. Every open device must be closed or output lands somewhere you did not choose — usually `Rplots.pdf`.

```
png(f, width, height) raster: units defaults to "px"; res is ignored unless units is physical
pdf(f, width, height) vector, for print. width/height are inches
svg(f, width, height) vector, for web. Needs a cairo build
jpeg(f, ..., tiff(...)) raster, lossy / lossless
dev.off() close the newest device; returns its name and number
dev.cur() / dev.list() which device is active / which are open
dev.new() push a copy of the current device onto the stack
capabilities("cairo") TRUE if svg() will work
```

```
# dev.off() belongs INSIDE the loop, not after it
for (g in groups) {
  png(file.path(out, paste0(g, ".png")), width = 1800, res = 150)
  plot(x[[g]], y[[g]])
  dev.off()
}
```

PAR() — MARGINS AND TEXT

`mar` and `oma` are measured in **lines of text**, so they scale with `cex` and `pointsize`. A margin tuned on screen is wrong in a high-res export unless you raise `pointsize` too.

```
par(mar = c(b, l, t, r)) plot margins, in lines of text
par(oma = c(b, l, t, r)) outer margins, only for multi-panel figures
par(mgp = c(3, 1, 0)) margin lines, tick labels, axis title
par(las = 0, 3) tick labels: parallel, horizontal, perpendicular, vertical
par(bty = "n") box: "o", "t", "7", "u", " ", "f", "r"
par(xaxs = "i") "r" (default, 4% padding) or "i" (exact data range)
par(yaxs = "i") same, for the y axis
par(tck = -0.02) tick length; negative draws outside, 0 removes
par(cex = 1.2) scale all plot text
par(family = "mono") base font family
par(font.main = 2) point size, main text
las = 1 is worth setting on any chart with numeric ticks — it stops y-axis labels being drawn slanted.
```

PAR() — PANELS AND STATE

```
par(mfrow = c(2, 2)) panels filled by row
par(mfcol = c(2, 2)) panels filled by column
layout(matrix, widths, heights) tagged panels; widths/heights in 0–1
layout(1) cancel layout(), back to a single panel
par(new = TRUE) draw onto the current plot without clearing it
par(no.readonly = TRUE) capture all current settings, for restoring
op <- par(x) / par(op) set and capture in one call, then restore
par() is device state and persists until changed. Capture and restore rather than hand-writing old values back — and in a function, wrap the restore in on.exit() so an error cannot leave the device mangled.
```

```
init <- par(no.readonly = TRUE) # snapshot everything
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
plot(x1)
plot(x2)
par(init) # always restore
```

EXPORT RECIPES

```
units = "in", res = 300 the combination to use for anything printed
pdf(onefile = FALSE) one file per page — safer for long batch jobs
pdf(width = 6, height = 4) inches, not pixels
pointsize = 12 raise it on small or high-res exports, or labels vanish
file.info(f)$size print it in a batch loop; 0 means dev.off() was missed
```

COLOURBLIND-SAFE PALETTES (R >= 4.1)

```
palette.pals() # every built-in palette
palette.colors(palette = "Okabe-Ito") # 9 colours, colourblind-safe
palette.colors(palette = "Okabe-Ito", alpha = 0.5)
```

Use the hex values as **R prints them** — hand-typing them from memory is how palettes get subtly wrong. The first Okabe-Ito colour is black, so drop it with `[-1]` unless you want black as your first series.

```
# greyscale check: Rec. 601 luma, not the mean of the channels
rgb <- t(col2rgb(pal)) / 255
luma <- rgb %*% c(0.299, 0.587, 0.114)
grey <- rgb(luma, luma, luma)
```

If the greyscale version has series you cannot tell apart, colour was doing work it should not do alone. Add a second channel: `pch`, `lty`, or direct labels.

AXIS() — DRAWING YOUR OWN

```
xaxt = "n" / yaxt = "n" suppress the automatic axis first, or you draw two
axis(1, at, labels) 1 bottom · 2 left · 3 top · 4 right
axis(1, at, labels = FALSE) ticks with no labels
axis(4) add a second, right-hand axis — usually a different unit
axis(..., las = 1) force horizontal labels, e.g. for dates
outer = TRUE draw on the figure margin, not the plot region
Axis positions are plain numbers, so dates work by converting and formatting.
```

```
when <- as.Date("2026-01-01") + 0:6 * 30
plot(when, sales, type = "b", xaxt = "n")
axis(1, at = as.numeric(when), labels = format(when, "%b %d"), las = 1)
```

DRAWING PRIMITIVES

```
plot(x, y, type=) "p" points · "l" line · "b" both · "o" overplot · "h" histogram · "n" axes only
points() / lines() add to the current plot without clearing it
segments(x0,y0,x1,y1) straight segments: error bars, caps, reference lines
arrows(x0,y0,x1,y1) line with a head; args run from → to
polygon(x, y, col=) closed filled shape; CI bands via c(x, rev(x))
rect() / text() boxes and labels at coordinates
legend(x, y, legend) x/y coords, or "topleft", "bottomright", ...
text(x, y, labels) labels at coordinates; pos, offset, padj
mtext(text, side) margin text; side 1–4 as for axis()
abline(a, b) reference line; also v =, h =
```

STATE YOU MUST RESTORE

Every call below changes device state that outlives the call itself. If you change it, restore it — or the next plot in the session silently inherits it.

```
call leaves behind
par(mfrow / mfcol) multi-panel layout; layout(1) cancels it
par(mar/oma/mgp/las/bty) page geometry, persists until reset
par(new = TRUE) a later plot() draws on top instead of clearing
an open device output goes to the wrong place; dev.off() it
setwd() inside a script, the working directory stays changed
options() process-wide until changed back
```

Full explanations: Chapter 12, **Production Export & Graphical Devices** · viz-base.rsquaredacademy.com